HSU, Ke-Cheng (Clio)

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Professional Summary

PhD candidate in Economics at HKUST. Analyzes how firms price, compete, and manage inventory, using large proprietary retail and e-commerce datasets and data collected through web scraping and APIs in Python. Builds testable models of demand, cost, and margins in Python, R, and C++, and writes the analysis up clearly for a non-technical reader. PhD coursework includes merger simulation, market entry and exit, and firm-conduct analysis. Seeking a Global Banking internship.

Education

The Hong Kong University of Science and Technology, Hong Kong

Doctor of Philosophy in Economics Expected Graduation: June 2027

Focus: firm pricing, competition, and industry analysis.

Master of Philosophy in Economics 2020–2022

CGA: 4.1/4.3.

Bachelor of Science in Mathematics and Economics, Minor in Actuarial Mathematics 2015–2019

CGA: 3.9/4.3.

Awards: Dean's List (2016–2019); University Scholarship (730K HKD); Academic Achievement Medal (top 1%); Postgraduate studentship (1.5M HKD); Redbird scholarship (40K HKD).

Experience

Research Assistant, National Taiwan University 2020

- Modeled crisis intervention in MATLAB as negotiation between a regulator and bank creditors; a voluntary bail-in occurs only when the threat not to bail out is credible.
- Showed that sparser interbank networks make that threat more credible in large shocks and reduce free-riding among contributing creditors.

Teaching Assistant & Lecturer, Department of Economics, HKUST 2020–present

- Lecturer for MSc Math Camp (4.53/5); TA for PhD and undergraduate economics (avg 4.72/5).

Projects

Pricing and Inventory in Japanese Book Retail *(w/ K. Kawaguchi, HKUST; K. Kawai, U. Tokyo)*

- Built a dynamic model of pricing and inventory for a retail oligopoly under resale price maintenance and uncertain demand, including stockouts and unsold returns (35–40% of books printed are discarded).
- Constructed title-by-market-by-month panels from bookstore point-of-sale data (deliveries, sales, returns; tens of billions of records across 1,000+ stores), linked by ISBN and merged with local demographics and title-level information scraped from Amazon in Python.
- Estimated demand and cost with instrumental variables (2SLS) and Hausman-style price instruments; checked the estimator with Monte Carlo; selected controls with LASSO and panel fixed effects; coded the equilibrium iteration in C++. Ignoring inventory and uncertain demand overstates marginal cost by about 20% and would misstate margins.

Pricing and Returns in Apparel E-Commerce *(Taobao, 2017–2023)*

- Collected product attributes in Python: wrote a scraping library, connected to an API, and used OCR to convert product images into text.
- Cleaned raw Taobao exports of orders, products, delivery, insurance, and addresses into linked transaction tables (2017–2023; ∼62M RMB; six apparel categories; return rates ∼20–30%): dropped non-products, giveaways, partial refunds, and orders missing logistics; separated retailer-provided from consumer-purchased insurance; standardized addresses to provinces.
- Estimated how purchases and returns respond to price and insurance via GMM (Python, R, C++ on AWS and HPC). Broader insurance expands purchases where margins are high; selective adoption protects seller profit where margins are thin by cutting returns.

Skills

Languages: Mandarin (Native), English (Fluent), Cantonese (Basic).

Technical: Python, R, MATLAB, Stata, C++, LaTeX, AWS, HPC/SLURM; web scraping, APIs, OCR; instrumental variables, maximum likelihood estimation, generalized method of moments, LASSO, bootstrap, Monte Carlo simulation, nonlinear optimization.

References: Kei Kawai (UTokyo; Compass Lexecon), Kohei Kawaguchi (HKUST; UTokyo Econ Consulting), Pak Hung Au (HKUST).